Jules Berman

New York, NY. +1 917-209-8726
[julesmichaelberman@gmail.com] ▪ [website] ▪ [Google Scholar]

Summary

PhD candidate building probabilistic and generative models for data-driven and PDE-based physics simulation. Author on **5 NeurIPS/ICML/ICLR papers** and **Harold Grad Prize winner** (top PhD '24). My work spans:

- **neural reduced order models that accelerate physics simulation** by >100× for parametric PDEs
- **time series modeling** — predictive approaches using CNNs, transformers, state-space models
- **models for stochastic simulation**—new generative approaches based on dynamic transport methods

Education

Ph.D., Computer Science — New York University, Courant Institute. GPA 3.9. Expected May 2026
Awards: Harold Grad Memorial Prize (top PhD student, 2024) · MacCracken Fellowship (full award)
Advisor: Benjamin Peherstorfer

B.S., Computer Science — New York University. May 2017
Awards: cum laude

Select Publications

Parametric model reduction of [...] stochastic systems via higher-order action matching **NeurIPS 2024**
`[arxiv]` **J. Berman**, T. Blickhan, B. Peherstorfer.

CoLoRA: Continuous low-rank adaptation for reduced neural modeling [...] **ICML 2024**
`[arxiv]` **J. Berman**, B. Peherstorfer.

Randomized Sparse Neural Galerkin Schemes with Deep Networks ***spotlight*** ▪ **NeurIPS 2023**
`[arxiv]` **J. Berman**, B. Peherstorfer.

Stochastic Lifting for One-Step Per-Frame Video Generation and Physics Simulation ***in review*** ▪ **ICLR 2026**
`[arxiv]` **J. Berman**, T. Blickhan, B. Peherstorfer.

Experience

Google May 2025 – Aug 2025
Research Engineer Intern
Developed approach to generating time series embeddings via masked autoregression using large scale transformers; cut downstream model size 3–5× while improving accuracy by 10–25% across datasets.

Meta May 2024 – Aug 2024
Research Scientist Intern, CTRL-Labs
Designed and trained generative diffusion models of EMG data, improving downstream gesture classification accuracy by 10% through synthetic data augmentation.

Flatiron Institute May 2021 – Aug 2022
Research Analyst, Center for Computational Neuroscience
Architected a distributed training platform for large-scale experiments (100k+ VAEs) and developed novel representational metrics enabling comparative analyses of latent representations.

Bloomberg LP April 2018 – April 2021
Senior Software Engineer, Global Infrastructure Team
Deployed ML-based predictive systems to forecast future infrastructure demands from historical data.

Software: Python · JAX · PyTorch · Optax · Flax · NumPy · SLURM · Git · LaTeX
Projects: Developed an online free-to-play poker web application `[justpoker.games]`
Service: Section leader for Optimization and Computation Linear Algebra; Reviewer at NeurIPS, ICLR, ICML;